Chapter 4
Proofs with structure, II

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What you should be able to do

- Read nested quantifiers in the order in which they bind variables.
- Prove both directions of an equivalence.
- Separate the existence and uniqueness obligations in a claim.
- Eliminate an impossible case and prove a negative statement.
- Match each logical step to a small Lean tactic.

Working convention

Domains matter: $\mathbb{N} = \{0, 1, 2, \ldots\}$; $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ have different possible witnesses. Lean snippets use the textbook's educational tactic environment.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 4, Sections 4.1–4.5.

4.1: A universal statement is a reusable rule

Statement	To prove it	To use it
$\forall x \in A,\ P(x)$	Fix an arbitrary $x \in A$.	Substitute a particular $t \in A$.
$P \to Q$	Assume P; prove Q.	Supply a proof of P to obtain Q.

$$\forall x \in \mathbb{R}, \quad x^2 \leq 2 \longrightarrow x \leq a$$

This rule only supplies $t \leq a$ after we establish $t^2 \leq 2$.

Lean connection

`intro x hx` introduces the object and its assumption. `exact h t ht` specializes a rule; `apply h` works backward.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1.

A best lower bound: choosing versus fixing a variable

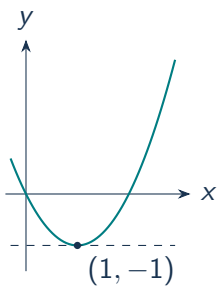
If $a \leq x^2 - 2x$ for every real x, use $x = 1$:

$$a \leq 1^2 - 2 \cdot 1 = -1.$$

Conversely, -1 is a lower bound because for an arbitrary $x \in \mathbb{R}$,

$$x^2 - 2x = (x-1)^2 - 1 \geq -1.$$

Thus the largest lower bound is -1.



Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1, Examples 4.1.1 and 4.1.5.

Quantifiers determine the order of the proof

Goal: $\exists b \in \mathbb{R},\ \forall x \in \mathbb{R},\ b \leq x^2 - 2x$.

```
-- Demonstration in the textbook's tactic environment.
example : Exists (fun b : Real =>
    forall x : Real, b <= x ^ 2 - 2 * x) := by
  use -1
  intro x
  calc
    -1 <= -1 + (x - 1) ^ 2 := by extra
    _ = x ^ 2 - 2 * x := by ring
```

The witness b is chosen *before* x is introduced. It cannot depend on x.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1, Example 4.1.5; ASCII teaching adaptation.

A quantified hypothesis can prove uniqueness

Suppose $a, b \in \mathbb{R}$ satisfy

$$a^2 \leq 2, \quad b^2 \leq 2, \qquad \forall y \in \mathbb{R},\ y^2 \leq 2 \Rightarrow y \leq a,$$

$$\forall y \in \mathbb{R},\ y^2 \leq 2 \Rightarrow y \leq b.$$

1. Apply the rule for b to $y = a$, using $a^2 \leq 2$: obtain $a \leq b$.
2. Apply the rule for a to $y = b$, using $b^2 \leq 2$: obtain $b \leq a$.
3. Antisymmetry of $\leq$ gives $a = b$.

Lean connection

`apply le_antisymm` produces the two inequalities. Each one is an application of a quantified implication.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1, Example 4.1.4.

“For all sufficiently large” has a quantifier order

For integers, this means

$$\exists N \in \mathbb{Z},\ \forall n \in \mathbb{Z}, \quad N \leq n \Rightarrow P(n).$$

For example, take $N = 5$ to show $n^3 \geq 4n^2 + 7$ eventually. For any integer $n \geq 5$,

$$n^3 - 4n^2 = n^2(n-4) \geq n^2 \geq 25 \geq 7.$$

The first inequality uses $n - 4 \geq 1$ and $n^2 \geq 0$.

What the proof supplies

A single threshold works for every later input. It need not be the smallest possible threshold.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1, Example 4.1.7.

4.1 Practice: identify the next object or assumption

1. Let $a \in \mathbb{Q}$ and suppose $a \geq -3 + 4b - b^2$ for every $b \in \mathbb{Q}$. Prove $a \geq 1$.
2. Prove $\exists n \in \mathbb{N},\ \forall m \in \mathbb{N},\ n \leq m$.
3. Prove $\exists a \in \mathbb{R},\ \forall b \in \mathbb{R},\ \exists c \in \mathbb{R},\ a + b < c$.

For each proof, state which witnesses may depend on previously introduced variables.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1, Exercises 4.1.10.

4.1 Solution ideas

1. Specialize the hypothesis at $b = 2$:

$$a \geq -3 + 4 \cdot 2 - 2^2 = 1.$$

2. Choose $n = 0$. Every natural number m satisfies $0 \leq m$.
3. Choose $a = 0$. After fixing an arbitrary b, choose $c = b + 1$. Then $a + b = b < b + 1 = c$.

Dependency check

In the third proof, a is independent of b; c is allowed to depend on b. Reversing quantifiers can change the problem.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1, Exercises 4.1.10.

4.2: An equivalence has two obligations

$$P \leftrightarrow Q \quad \text{means} \quad (P \to Q) \wedge (Q \to P).$$

For $a \in \mathbb{Q}$,

$$3a + 1 \leq 7 \quad \Longleftrightarrow \quad a \leq 2.$$

Forward: subtract 1 and divide by positive 3. Backward: multiply by positive 3 and add 1.

```
-- Demonstration: each branch has its own assumption.
example (P Q : Prop) (forward : P -> Q)
    (backward : Q -> P) : Iff P Q := by
  constructor
  . exact forward
  . exact backward
```

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.2, Example 4.2.1; adapted tactic demonstration.

Divisibility equivalence: the difficult direction needs an idea

For $n \in \mathbb{Z}$, prove $8 \mid 5n \iff 8 \mid n$.

Forward direction

Write $5n = 8a$ with $a \in \mathbb{Z}$. The identity $-3 \cdot 5 + 2 \cdot 8 = 1$ gives

$$n = -3(5n) + 16n = 8(-3a + 2n).$$

The integer $-3a + 2n$ is the divisibility witness.

Backward direction

If $n = 8a$, then $5n = 8(5a)$.

Do not cancel a factor in a divisibility statement without a reason; the coprimality of 5 and 8 is doing the work.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.2, Example 4.2.2.

Solving an equation is an equivalence proof

For a real number x,

$$x^2 + x - 6 = 0 \quad \iff \quad x = -3 \ \vee \ x = 2.$$

1. If the equation holds, then $(x + 3)(x - 2) = 0$.
2. The zero-product rule gives $x + 3 = 0$ or $x - 2 = 0$.
3. These equations give the two candidate roots.
4. Conversely, substitution of either -3 or 2 makes the original polynomial zero.

Lean connection

`rw [mul_eq_zero] at h` turns a zero-product hypothesis into a disjunction. Use `obtain h1 | h2 := h` to handle its branches.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.2, Examples 4.2.5 and 4.2.7.

4.2 Practice: prove the converse too

1. For $x \in \mathbb{R}$, prove $2x - 1 = 11 \iff x = 6$.
2. For $k \in \mathbb{N}$, prove

$$k^2 \leq 6 \quad \iff \quad k = 0 \ \vee \ k = 1 \ \vee \ k = 2.$$

3. Explain why item 2 is false if $k \in \mathbb{R}$.

Label the forward and backward directions before calculating.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.2, Exercises 4.2.10; domain check is an extension.

4.2 Solution ideas

1. Forward: $2x = 12$, so $x = 6$. Backward: $2 \cdot 6 - 1 = 11$.
2. If $k^2 \leq 6$, then $k \geq 3$ would imply $k^2 \geq 9$, a contradiction. Thus $0 \leq k \leq 2$, giving the three integer cases. Conversely, their squares are $0, 1, 4$, all at most 6.
3. Over $\mathbb{R}$, $k = \frac{1}{2}$ is a counterexample to the forward implication.

Proof habit

A finite list follows from bounds only in a discrete domain. For an equivalence, checking the listed values proves only one direction.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.2, Exercises 4.2.10.

4.3: Existence plus uniqueness

$$\exists! x \in A,\ P(x) \quad \iff \quad \exists x \in A,\ P(x) \wedge \forall y \in A, (P(y) \to y = x).$$

To prove $\exists! a \in \mathbb{R},\ 3a + 1 = 7$:

1. **Choose:** $a = 2$.
2. **Verify existence:** $3 \cdot 2 + 1 = 7$.
3. **Verify uniqueness:** for any y with $3y + 1 = 7$,

$$y = \frac{(3y + 1) - 1}{3} = \frac{7 - 1}{3} = 2.$$

`use 2`, then `constructor`, separates the two obligations in Lean.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.3, Example 4.3.1.

Use uniqueness by building a second candidate

Let $x \in \mathbb{Q}$. Suppose exactly one rational a satisfies $a^2 = x$.

1. Existence supplies a rational a with $a^2 = x$.
2. The number $-a$ is another candidate: $(-a)^2 = a^2 = x$.
3. Uniqueness forces $-a = a$.
4. Therefore $2a = 0$, so $a = 0$ and $x = a^2 = 0$.

Why existence matters

A rational number such as -1 has no rational square root. "At most one" would hold for it, but the conclusion $x = 0$ would fail.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.3, Example 4.3.3.

A unique representative modulo 5

Prove $\exists! r \in \mathbb{Z}$ such that $0 \leq r < 5$ and $14 \equiv r \pmod 5$.

Existence

Take $r = 4$: $14 - 4 = 5 \cdot 2$ and $0 \leq 4 < 5$.

Uniqueness

If $0 \leq r < 5$ and $14 - r = 5q$, then

$$10 \leq 14 - r \leq 14, \qquad 2 \leq q < 3.$$

Since q is an integer, $q = 2$; hence $r = 14 - 10 = 4$.

Without the interval restriction, $4, 9, -1, \ldots$ all represent the same congruence class.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.3, Example 4.3.4.

4.3 Practice: do not stop after finding a witness

1. Prove $\exists! x \in \mathbb{Q},\ 4x - 3 = 9$.
2. Prove $\exists! n \in \mathbb{N},\ \forall a \in \mathbb{N},\ n \leq a$.

For each item, write two separate sentences:

- The chosen object has the required property because ...
- Every other object with the property equals it because ...

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.3, Exercises 4.3.5.

4.3 Solution ideas

1. Choose $x = 3$ and check $4 \cdot 3 - 3 = 9$. If $4y - 3 = 9$, then $4y = 12$ and $y = 3$.
2. Choose $n = 0$ and use $0 \leq a$ for each $a \in \mathbb{N}$. If $m \leq a$ for every natural a, specialize at $a = 0$. Then $m \leq 0$, and since $0 \leq m$, we obtain $m = 0$.

Reusable pattern

A universal property can identify a unique object by testing it at a strategically chosen input.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.3, Exercises 4.3.5.

4.4: An impossible case imposes no further obligation

If the assumptions give both P and $\neg P$, they cannot all hold. In logic, from this contradiction we may derive any target Q.

```
-- Demonstration of the logical rule.
example (P Q : Prop) (hP : P) (hnP : Not P) : Q := by
  contradiction
```

For example, $t \in \mathbb{Z}$, $t < 3$, and $t - 1 = 6$ imply both $t = 7$ and $7 < 3$.

Interpretation

The conditional theorem is valid because no input satisfies its assumptions. This is useful when eliminating a branch of a case split.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.4, Examples 4.4.1 and 4.4.2.

Rule out the wrong sign

Let $x, y \in \mathbb{R}$, $x \geq 0$, and $xy > 0$. Prove $y > 0$.

1. Split into $y \leq 0$ or $y > 0$.
2. In the first case, multiplying by $x \geq 0$ gives

$$xy \leq x \cdot 0 = 0,$$

   contradicting $xy > 0$.
3. In the second case, the desired conclusion is already available.

Where the hypothesis enters

The condition $x \geq 0$ justifies the direction of the inequality. Without it, $x = y = -1$ would be a counterexample.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.4, Example 4.4.1.

Residue cases can expose a contradiction

For $n \in \mathbb{Z}$, suppose $n^2 + n + 1 \equiv 1 \pmod 3$.

$n \pmod 3$	$n^2 + n + 1 \pmod 3$	Consequence
0	1	Allowed
1	0	Contradicts the assumption
2	1	Allowed

Thus $n \equiv 0 \pmod 3$ or $n \equiv 2 \pmod 3$.

Lean connection

`mod_cases` creates the residue cases. The excluded case yields $0 \equiv 1 \pmod 3$; the textbook uses `numbers at H` on that numeric contradiction.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.4, Example 4.4.3.

4.4 Practice: eliminate impossible branches

1. Let $x \in \mathbb{Q}$, $x^2 = 4$, and $x > 1$. Prove $x = 2$ by factoring and considering cases.
2. Let $n \in \mathbb{Z}$ and $n^2 \equiv 4 \pmod 5$. Prove

$$n \equiv 2 \pmod 5 \quad \vee \quad n \equiv 3 \pmod 5.$$

For each excluded case, state the contradictory pair of facts explicitly.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.4, Exercises 4.4.6.

4.4 Solution ideas

1. $(x+2)(x-2)=0$, so $x=-2$ or $x=2$. The first case gives $1<-2$, contradicting the numerical order. Therefore $x=2$.
2. Square the five possible residues:

$n \pmod 5$	0	1	2	3	4
$n^2 \pmod 5$	0	1	4	4	1

   Only residues 2 and 3 match the given square residue 4.

A case split is complete only when every branch is either solved or contradicted.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.4, Exercises 4.4.6.

4.5: To prove a negation, assume its target

The logical meaning of $\neg P$ is $P \to \mathrm{False}$. To prove $\neg \forall x \in \mathbb{R},\ x^2 \geq x$, assume that universal claim and test $x = \frac{1}{2}$.

$$\left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4} < \tfrac{1}{2}.$$

```
-- Demonstration in the textbook's tactic environment.
example : Not (forall x : Real, x ^ 2 >= x) := by
  intro h
  have H := h (1 / 2)
  numbers at H
```

The newly introduced assumption is discharged by the proof of the negation.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.5, Example 4.5.1.

Why a number between consecutive multiples is not divisible

For integers a, b, q, assume $bq < a < b(q+1)$. Then $b \nmid a$.

1. The interval is nonempty, so $bq < b(q+1)$ and hence $b > 0$.
2. If $a = bk$ for some integer k, then

$$bq < bk < b(q+1) \quad \Rightarrow \quad q < k < q+1.$$

3. There is no integer strictly between consecutive integers.

Example: $3 \cdot 4 < 13 < 3 \cdot 5$, so $3 \nmid 13$.

Lean connection

`intro h` assumes divisibility; `obtain` extracts its integer witness. The contradiction is an impossible bound on that witness.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.5, Examples 4.5.2 and 4.5.8.

A contradiction-based test for primality

Definition

A natural number p is prime when $p \geq 2$ and every natural divisor of p equals 1 or p.

If $p = kl$ for $k, l \in \mathbb{N}$ with $k \neq 1$ and $k \neq p$, then p is not prime: $k \mid p$, but neither allowed alternative holds. Equivalently, to prove p prime, it suffices to show

$$p \geq 2, \qquad \forall m \in \mathbb{N},\ 1 < m < p \Rightarrow m \nmid p.$$

A divisor of positive p lies between 1 and p. Eliminating the interior leaves exactly the two endpoints.

Source: Heather Macbeth, *The Mechanics of Proof*, Sections 4.1, 4.4, 4.5; Examples 4.1.8, 4.4.4, 4.5.7.

A smaller search interval for a prime test

Let $p, T \in \mathbb{N}$. Assume

$$p \geq 2, \qquad p < T^2, \qquad \forall m \in \mathbb{N},\ 1 < m < T \Rightarrow m \nmid p.$$

Then p is prime.

1. If p were composite, write $p = ml$ with $1 < m < p$.
2. Since $ml = p > m$, we have $l > 1$.
3. If both $m \geq T$ and $l \geq T$, then $p = ml \geq T^2$, a contradiction.
4. One factor lies strictly between 1 and T, contrary to the assumption.

For $p = 79$, use $T = 9$ and check divisors $2, 3, \ldots, 8$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.5, Example 4.5.9.

4.5 Practice: plan the contradiction before calculating

1. Prove that no natural number n satisfies $n^2 = 2$.
2. Use the improved prime test with $T = 10$ to prove 97 prime. For each $m = 2, \ldots, 9$, locate 97 between consecutive multiples of m.

For item 1, split the natural numbers at the boundary between 1 and 2.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.5, Example 4.5.4 and Exercises 4.5.10.

4.5 Solution ideas

1. If $n \leq 1$, then $n^2 \leq 1 < 2$. If $n \geq 2$, then $n^2 \geq 4 > 2$. Both cases contradict $n^2 = 2$.
2. We have $2 \leq 97 < 10^2$. The lower quotients are:

m	2	3	4	5	6	7	8	9
q	48	32	24	19	16	13	12	10
mq	96	96	96	95	96	91	96	90

In every column, $mq < 97 < m(q+1)$, so $m \nmid 97$. The improved prime test now applies.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.5, Example 4.5.4 and Exercises 4.5.10.

Review: let the outermost logical symbol guide you

Goal	Mathematical action	Lean cue
$\forall x,\ P(x)$	Fix an arbitrary x	`intro x`
$P \to Q$	Assume P	`intro hP`
$P \leftrightarrow Q$	Prove two implications	`constructor`
$\exists! x,\ P(x)$	Witness, property, uniqueness	`use`, `constructor`
$\neg P$	Assume P and derive falsehood	`intro hP`
Any goal with $P, \neg P$	Eliminate the impossible case	`contradiction`

Before leaving a proof, check domains, witness dependencies, both directions, and all cases.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 4, Sections 4.1–4.5.

Sources and further work

Heather Macbeth, *The Mechanics of Proof*, Chapter 4.

- 4.1: universal quantification, implication, and eventual properties.
- 4.2: equivalence and complete solution sets.
- 4.3: existence and uniqueness.
- 4.4–4.5: contradictions, nonexistence, and prime tests.

Read Chapter 4 and its remaining exercises.

These slides use original teaching prose and drawings. Example and exercise numbers in source comments identify the adaptation. Lean demonstrations use the textbook's educational tactics.